A Novel Method Combining Global Visual Features and Local Structural Features for SAR ATR

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***Abstract*— The mainstream synthetic aperture radar (SAR) automatic target recognition (ATR) methods commonly use convolutional neural network (CNN) to extract the visual information of SAR targets, while the physical structural information is seldom considered. Scattering center features can describe the targets' physical structure information and are robust to the local variations of targets, which can be exploited to reflect the local structural characteristic of SAR targets. Therefore, we propose a novel method that effectively combines global visual features and local structural features for SAR ATR. The local structural features here contain not only the local physical structure information but also the local visual information. Our proposed method consists of three parts: global-based module, local-based module, and feature fusion module. Global-based module utilizes CNN to extract global visual features from SAR images. Local-based module first extracts attributed scattering centers (ASCs) from the complex SAR image and models each ASC as a node to construct graph data, from which we further use a multiscale graph convolutional network (GCN) to extract local structural features. The node features for GCN learning are constructed by multiplying the corresponding local visual features in shallow CNN feature maps with the ASC reconstruction maps to better reflect the local structure characteristic. Then, the learned local structural features from GCN are further fused with global visual features to achieve SAR ATR. As far as authors know, this is the first work combining CNN and GCN to effectively extract global visual features and local structural features simultaneously in SAR ATR. Experiments on the moving and stationary target acquisition and recognition (MSTAR) dataset show that our proposed method outperforms SOTA methods in terms of classification accuracy.**

***Index Terms*— Attributed scattering center (ASC), convolutional neural network (CNN), graph convolutional network (GCN), synthetic aperture radar (SAR), target recognition.**

I. Introduction

As an active imaging system, synthetic aperture radar (SAR) holds the advantages of all-weather and full-time conditions, which have been widely applied in both military and civilian fields [1]. SAR automatic target recognition (ATR) provides the key information for SAR image interpretation and has attracted extensive research and attention [2], [3]. However, due to the characteristics of its microwave band imaging mechanism and phase coherent processing, the visual appearance of SAR images is notably different from optical images, which makes it a difficult task to interpret the targets. Meanwhile, the inherent SAR speckle noise makes it more sensible and difficult to extract useful information for SAR target recognition.

As the most widely used deep networks, convolutional neural networks (CNNs) can automatically extract visual information and have been extensively utilized in SAR ATR. Chen et al. [4] used convolutional layers instead of fully connected layers, while the physical structure information of SAR targets is neglected. The physical structure of SAR targets is relatively stable with local variations; thus, the physical structure of SAR targets is significant for SAR ATR. Different from the visual features extracted from the amplitude information, scattering center features [5], [6], [8] are commonly used for characterizing the electromagnetic scattering characteristics and the distribution can describe the physical structure. Zhang et al. [5] utilized scattering center features and employed a sparse reconstruction method for target recognition. Several latest studies [6], [7], [8] combined the scattering center features with CNN features to sufficiently utilize the information of SAR targets. Zhou et al. [6] fused CNN and attributed scattering center (ASC) schematic map to improve the performance of SAR ATR. Li et al. [7] used component decomposition to extract component features via ASCs and then fused them with the CNN features. Zhang et al. [8] proposed a novel method FEC, which combines CNN and ASC features for recognition. Although those methods [6], [7], [8] integrated the ASC information into CNN, they treated the extracted ASCs of an SAR image as multiple components or a whole and the distribution relationship of ASCs is neglected. Our previous work [15] combined scattering centers with GCN to utilize physical structural information while the more accurate ASC information is not used. Based on the above analysis, exploring effective methods using ASC information to extract physical structural features to better utilize the electromagnetic scattering characteristic of SAR targets is still worthy of study.

Graph convolutional networks (GCNs) [9] and transformers have obtained enormous progress in recent years. The traditional CNN is applied to grid-such as data such as images while fails on graph-structured data. GCN is an extension of CNN, which is proposed to operate convolution on graph data and is more suitable for extracting relation and structure information of the data. Some latest researches [10], [11], [12] have introduced GCNs and transformers to tackle hyperspectral image classification and remote sensing scene classification. Hong et al. [10] developed a new minibatch GCN (miniGCN) and fused it with CNN for hyperspectral image classification.

Manuscript received 1 August 2023; revised 31 August 2023; accepted 13 September 2023. Date of publication 13 October 2023; date of current version 30 October 2023. This work was supported in part by the National Science Foundation of China under Grant U21B2039. *(Corresponding author: Lan Du.)*

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Digital Object Identifier 10.1109/LGRS.2023.3323655

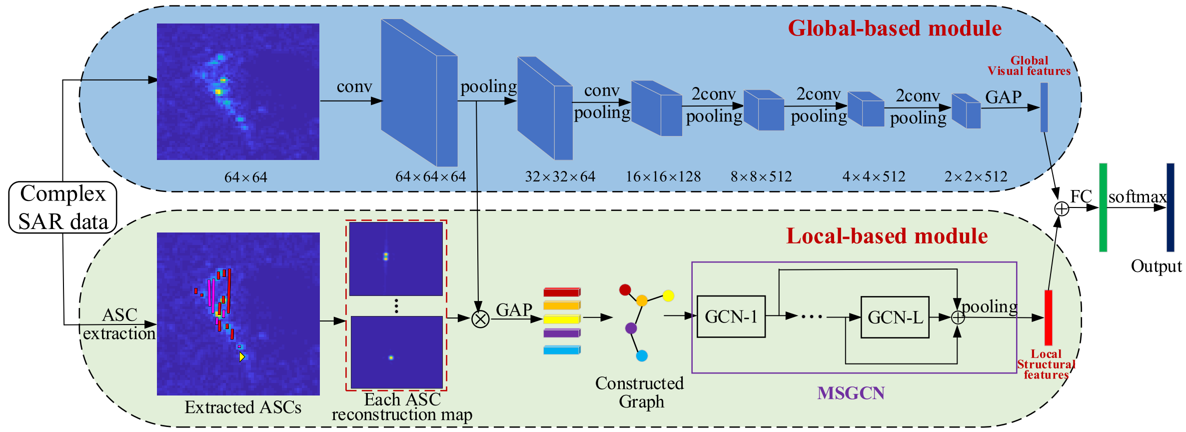


Fig. 1. Architecture of our proposed method. The top module uses CNN to extract shallow and deep visual features. The bottom module uses multiscale GCN to extract local structural features from the constructed graph, which fuse the shallow CNN features and the ASC reconstruction maps. The deep CNN features and the local structural features extracted by MSGCN are then fused by the concatenation operation and delivered to the softmax classifier.

Hong et al. [11] proposed a SpectralTransformer to focus on extracting spectral information. Liang et al. [12] used GCN to investigate the dependencies among objects and fused them with CNN for remote sensing scene classification. All those methods consider the spatial relationship of targets in hyperspectral images or remote images, while they do not investigate the relationship of SAR images. The existing GCN-based methods for SAR ATR, such as [13] and [14], model each image as a node to explore the similarity relationship of nodes to classify images, while the structural information which is inherently contained in SAR target is ignored and the electromagnetic scattering characteristic is not utilized. Li et al. [15] combined the scattering centers with GCN to effectively extract the structural features while the visual features are not adequately considered. The existing methods either fuse the visual features and the ASC features [6], [7], [8] or merely use GCN [15] to extract the structural features, while they do not fully consider both visual features and structural features for SAR ATR. Therefore, how to effectively extract structural features of the SAR target via GCN and further combine the structural features with the visual features is significant for SAR ATR.

Inspired by our previous work [15], we propose a novel method combining CNN and multiscale GCN for SAR ATR, which is denoted as CNN-multi-scale GCN (MSGCN) to extract global visual features and local structural features simultaneously. The local structural features here contain not only the local physical structure information but also the local visual information. Considering that only relying on the extracted ASC parameters is insufficient for GCN learning, the node features are constructed by multiplying the corresponding local visual features of each ASC in shallow CNN feature maps with the ASC reconstruction maps, which are further inputted into GCN to learn the local structural features. Then, the learned local structural features are fused with global visual features.

The novelty of our proposed method is shown as follows.

1) As far as our knowledge, this is the first research to integrate the ASC features into GCN to utilize the physical structural information for SAR ATR.
2) We design an effective fusion strategy to fuse global visual features and local structural features via the combination of CNN and GCN for SAR ATR. The local structural features here contain not only the local physical structural information but also the local visual information. The node features for GCN learning are constructed by multiplying the corresponding local visual features in shallow CNN feature maps with the ASC reconstruction maps to better reflect the local structure characteristic.
3) Compared with the existing SOTA methods, our proposed method more sufficiently utilizes the information of SAR targets and achieves superior performance.

II. Methodology

The overall architecture of our proposed method is illustrated in Fig. 1, which includes the global-based module, local-based module, and feature fusion module. The global-based module designed a modified VGG16 network to obtain visual features from the real-valued images. For the local-based module, ASC sets are first extracted from each SAR target; then, the ASC reconstruction maps are obtained according to the imaging mechanism. The ASC reconstruction maps are fused with the shallow CNN features to extract individual features for each ASC, which are then converted into a graph by modeling each ASC as a node to sufficiently reflect the physical structure information of SAR targets. To learn different levels of neighbor information from the constructed graph, MSGCN is proposed to fuse the multilayer GCN features for extracting structural features. The fusion in MSGCN can enhance the feature representability of structural features. Finally, the global visual features and the structural features extracted by MSGCN are fused. The three modules are elaborated in Sections II-A–II-C.

A. Global-Based Module

The global-based module is illustrated in the top module in Fig. 1, which is modified from VGG16 [14]. Considering that the acquired SAR target data are limited, the global-based module cuts out the original VGG16 network to prevent overfitting. Specifically, the global-based module is assembled with eight convolution blocks and five max-pooling operations to generate the visual feature maps. The size of convolutional kernels is set as 3×3. The global visual features are obtained via a global average pooling (GAP) on the final CNN feature maps and are subsequently used to fuse with the local structural features based on the output of the local-based module and then achieve target recognition through the classifier.

B. Local-Based Module

First, we extract ASCs in an SAR image through the ASC extraction method in [17]. Second, we convert the extracted ASCs into a graph. Third, the local structural features are extracted by MSGCN from the constructed graph.

1) ASC Extraction: The radar echoes of SAR system can be approximated as a superposition of multiple ASCs [18]

$$\mathbf{E}(f,\varphi)=\sum_{i=1}^{K}E_i(f,\varphi) \tag{1}$$

where f and φ represent the frequency and aspect angle, respectively, and K is the number of the ASCs. An individual ASC obeys the ASC model as follows:

$$E_i(f,\varphi)=S_i\cdot\left(j\frac{f}{f_c}\right)^{\alpha_i}\cdot\exp\left(\frac{-j4\pi f}{c}(x_i\cos\varphi+y_i\sin\varphi)\right)\cdot\sin c\left(\frac{2\pi f}{c}L_i\sin\left(\varphi-\bar{\varphi}_i\right)\right)\cdot\exp(-2\pi f\gamma_i\sin\varphi) \tag{2}$$

where c represents the velocity of the electromagnetic wave, $\boldsymbol{\theta}=\{\boldsymbol{\theta}_i\}=[S_i,x_i,y_i,\alpha_i,\gamma_i,L_i,\bar{\varphi}_i](i=1,2,\ldots,K)$ represents the attribute sets of all the ASCs in an SAR image, S_i is the complex amplitude, x_i and y_i are the range position and aspect position, respectively, α_i and γ_i denote the frequency-dependent factor and the aspect-dependent factor, respectively, L_i is the length of the distributed ASC, and $\bar{\varphi}_i$ is the initial aspect angle. The parameter estimation of ASCs is realized using the extraction algorithm based on image-domain sparse representation [17].

2) Graph Construction: After extracting the ASC sets of SAR targets, we model each ASC as a node in a graph and establish the connection of nodes to represent the dependency relationship among ASCs. Since the node features are critical in GCN learning and only relying on the extracted ASC parameters is insufficient, we multiply the corresponding local visual features of each ASC in shallow CNN feature maps with the ASC reconstruction maps to enhance the local structural characteristic. The combined features are converted into feature vectors by GAP, which acts as the initial node features of the constructed graph.

Inspired by the graph construction in [15], the dependency of different ASCs can be modeled using the corresponding position parameter $\boldsymbol{m}_i=[x_i\,y_i]$; thus, the adjacency matrix can be formulated as follows:

$$a_{i,j}=\begin{cases}e^{\frac{\left\|m_i-m_j\right\|^2}{-2\gamma^2}}, & \text{if } \left\|\boldsymbol{m}_i-\boldsymbol{m}_j\right\|^2<\tau\\ 0, & \text{otherwise}\end{cases} \tag{3}$$

where parameter γ is the scalar factor of the Gaussian kernel and τ is the threshold of the distance between different ASCs.

3) MSGCN: Owing to the strong structure learning ability, the GCNs are adopted to extract local structural features from the constructed graph. Several GCN layers are generally stacked with each layer taking the node features h and the normalized adjacency matrix A as inputs. The node features describe the attributes of nodes and the adjacency matrix reflects the structure information. By using the graph convolution operation to aggregate the nodes features, the output node features of the first layer are formulated as follows:

$$h^{(1)}=\sigma(Ah^{(0)}W^{(0)}) \tag{4}$$

where σ is the nonlinear transformation function and $W^{(0)}$ represents the trainable parameters. Similar to stacking multiple convolutional layers in CNN, the high-level features of GCN can be represented as follows:

$$h^{(l)}=\sigma(Ah^{(l-1)}W^{(l-1)}) \tag{5}$$

where l denotes the number of GCN layers and $W^{(l-1)}$ is the corresponding trainable parameters at the lth layer. We concatenate the output features of different layers along the feature dimension to extract multiscale node features. Then, we use max-pooling as the readout operation on the multiscale node features and denote the results as the local structural features f_{str}.

C. Feature Fusion

Considering the global visual features extracted by CNN and the local structural features are complementary, we employ a feature fusion module to simultaneously learn both features for sufficiently utilizing the information of SAR targets. The fusion module can be represented as follows:

$$f_{\text{fus}}=\text{concat}(f_{\text{vis}},f_{\text{str}}) \tag{6}$$

where f_{vis} denotes the output of the global-based module and concat denotes the concatenation operation. The fused features are then inputted into a fully connected layer and output probability vector. The output node number of the fully connected layer is set to the category of the training dataset. The cross-entropy loss function is adopted as the optimization goal of our proposed CNN-MSGCN.

III. Experimental Results and Analysis

A. Datasets Description

The dataset used in this work is the measured moving and stationary target acquisition and recognition (MSTAR) dataset, which is the most widely used SAR ATR dataset. Ten categories of ground targets with a resolution of 0.3×0.3 m acquired in 1996 and 1997 are included in the dataset. The radar that acquired the data operates in the X-band and works in spotlight mode. We utilize the three-target MSTAR data and the ten-target MSTAR data [3], [5], [7], [15], which are two commonly used settings for SAR ATR in our experiments. The training and testing images on the ten-target data are shown in Table I. The three-target data include three categories (BMP2, BTR70, and T72), which are the same as those in the ten-target data.

TABLE I
TRAINING AND TESTING IMAGES FOR THE TEN-TARGET DATA

No.	Type		Training (17°)	Testing (15°)
1	BMP2	SN_C21	233	196
		SN_C9566	0	196
		SN_C9563	0	195
2	BTR70		233	196
3	T72	SN_132	232	196
		SN_S7	0	191
		SN_812	0	195
4	ZSU234		299	274
5	BTR60		255	195
6	ZIL131		299	274
7	BRDM2		298	274
8	T62		299	273
9	2S1		299	274
10	D7		299	274

TABLE II
ACCURACIES AND FLOPS OF DIFFERENT METHODS ON THE THREE-TARGET DATA AND TEN-TARGET DATA

Types	Input	Methods	Accuracy		FLOPs
			Three-target	Ten-target	
Traditional methods	Intensity image	Template matching	0.9311	0.8758	-
		DL-JDSR [3]	0.9391	0.9148	-
	Complex data	BCS [5]	0.9790	0.9260	-
Deep-learning-based methods	Intensity image	Euclidean-AE [2]	0.9414	0.9129	1.38×10^7
		VGGNet	0.9582	0.9322	1.13×10^9
		A-ConvNet	0.9684	0.9369	6.24×10^7
	Complex data	GCN [15]	0.9875	0.9838	1.43×10^9
	Intensity image, complex data	CNN-ASC [6]	0.9795	0.9869	1.07×10^9
		CA-MCNN [7]	0.9861	0.9781	9.45×10^8
		FEC [8]	-	0.9846	1.61×10^9
		Proposed method	**0.9912**	**0.9906**	$\mathbf{1.49\times10^9}$

B. Experimental Setup

In the three-target and the ten-target experiments, the original images with a size of 128×128 are clipped into 64×64 for reducing the computation cost; 20 ASCs are extracted for each SAR image via the image-domain-based sparse representation algorithm [17]. We use three GCN layers and set the output feature dimension of each layer as 512. ReLU is used as the nonlinear transformation function. We use the Adam optimizer to optimize the loss function. The initial learning rate is 0.01, and the learning rate decays by 0.5 every 100 epochs. The epoch number is set as 500. The PyTorch deep learning framework is used to implement the experiments.

C. Recognition Experiments

In this section, we conduct some experiments on the three-target and ten-target MSTAR data to validate the effectiveness of our proposed method. The comparison methods of SAR ATR include some traditional methods and several deep-learning-based methods. Dictionary learning and JDSR (DL-JDSR) [3], Bayesian compressing sensing (BCS) [5], and template matching method are as typical traditional methods for comparison. The deep learning-based methods consist of Euclidean distance restricted autoencoder [2] (Euclidean-AE), VGG CNN (VGGNet) [16], A-ConvNet [4], and some SOTA methods, such as CNN-ASC [6], multiscale CNN based on component analysis (CA-MCNN) [7], FEC [8], and GCN [15]. Methods [2], [4], [16] only take intensity SAR image as inputs and method [15] only takes complex SAR data as inputs. Both methods [6], [7], [8] and our proposed method take intensity image and complex data as inputs. No data augmentation strategy is used in all methods to ensure fairness. The recognition accuracies and computational complexity of the proposed CNN-MSGCN and the abovementioned compared methods on the three-target data and ten-target data are shown in Table II.

From Table II, we can see that our proposed method outperforms all the traditional methods and the deep-learning-based methods on both three-target data and ten-target data in terms of accuracy. For the three-target data, BCS achieves an accuracy of 0.9790, which outperforms other traditional methods. However, due to the weak feature extraction ability, traditional methods show relatively worse recognition performance than deep-learning-based methods. Among the deep-learning-based methods, Euclidean-AE [2], VGGNet, and A-ConvNet only use intensity image as the input of deep networks, which is insufficient for extracting the features, achieving lower accuracies than methods that use both intensity image and complex data. CNN-ASC [6], CA-MCNN [7], FEC [8], and our proposed method both consider intensity image and complex data, which utilize the information more sufficiently. Although methods [6], [7], [8] incorporate electromagnetic scattering information into deep networks, the mining of structural features is still limited. Our previous work [15] utilized GCN to extract structural features of SAR targets, while the visual features are not adequately considered. Our proposed method effectively combines the visual features and structural features via the fusion of CNN and GCN, better utilizes the information of SAR targets, and achieves superior recognition performance than the SOTA methods.

For the ten-target data, the recognition accuracy of our proposed method is also the highest compared with other methods, which further validates that our proposed method can still guarantee good recognition performance when the number of target categories increases. By the way, in terms of computational complexity, our proposed method does not have a significantly higher computation burden than those SOTA methods.

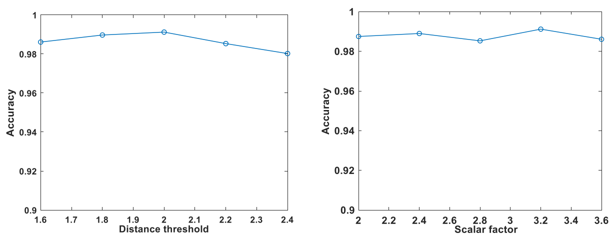
D. Ablation Experiments

In our proposed method, the designed CNN extracts global visual information and local visual information, which correspond to each extracted ASC, and the local-based module combines the local visual information and the local physical information to extract local structural features via MSGCN. To analyze the effect of different information utilization on performance, the ablation experiments on three-target data are given in Table III.

From row 1 and row 2 in Table III, we can see that the accuracy of using CNN and MSGCN separately is 0.9684

TABLE III
ABLATION EXPERIMENTS ON THE THREE-TARGET DATA

Global visual information	Local physical structure information	Local visual information	Accuracy
√	×	×	0.9684
×	√	×	0.8410
×	√	√	0.8681
√	√	×	0.9839
√	√	√	**0.9912**



(a) (b)

Fig. 2. Accuracies under different parameter settings. (a) Distance thresholds. (b) Scalar factors.

and 0.8410, respectively. By combining the local structural information and local visual information, a higher accuracy of 0.8681 is obtained compared with only utilizing the local structural information reflected by ASCs, validating the effectiveness of multiplying the local visual features with ASC reconstruction maps to enhance the local structural characteristic. We can see that directly fusing global visual information with local physical structural information can obtain an accuracy of 0.9839, which is higher than only using CNN or MSGCN, showing the combination of CNN and MSGCN is beneficial for recognition. Our proposed method combined local visual information with local physical information to extract local structural features and fuse them with global visual features, which achieves the highest accuracy, demonstrating the effectiveness of the proposed CNN-MSGCN.

E. Performance Under Different Parameter Settings

In the proposed method, the distance threshold and the scalar factor of the Gaussian kernel are critical for converting the extracted ASCs into graph-structured data. The recognition accuracies on three-target data vary with different parameter settings of both distance thresholds and scalar factors, which are given in Fig. 2.

It is obvious that the corresponding best values of parameters τ and γ are 2 and 3.2 according to the results in Fig. 2. When the distance threshold τ exceeds or is less than 2, the performance has an obvious drop, which means the constructed graph is too sparse or too strong to reflect the local structure information of SAR targets. Relatively speaking, the recognition performance is not very sensitive to the change in scalar factor of the Gaussian kernel.

IV. CONCLUSION

For efficiently and sufficiently leveraging the characteristic of SAR targets, we propose a unified method CNN-MSGCN to extract global visual features and local structural features simultaneously. We model each ASC as a node to convert the extracted ASCs into a graph for GCN learning. Then, the learned local structural features from GCN are further fused with global visual features extracted by CNN to obtain a robust feature representation for SAR ATR. The extensive experimental results on both MSTAR three-target data and ten-target data validate that our proposed method outperforms the existing methods in terms of recognition accuracy. The future direction of our work will extend our GCN-based method to dynamic structure learning and enable our model adaptative to other extending operation conditions in practical SAR ATR.

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